

CONTROLLED UNCLASSIFIED INFORMATION

Justification and Approval (J&A) for Other Than Full and Open Competition

Was a J&A approved for the preceding acquisition where that acquisition required a J&A? ☒ Yes ☐ No

!! Attach the preceding J&A in the staff package for this J&A. The preceding J&A will be used as a reference document.

Is this a new or amended J&A Document? ☒ New ☐ Amended (Prior to Award Only!)

Is this a Bridge Action as defined at DAFFARS 5302.101? ☐ Yes ☒ No

Dollar Value of this Acquisition: ☐ ≤ \$750K ☐ > \$750K and ≤ \$15M ☐ > \$15M and ≤ \$100M ☒ > \$100M

Contracting Activity: Air Force Installation Contracting Center (AFICC)

Purchase Request (if available) / Local ID Number: TBD

Program / Project (and PE, if applicable): HVAC Chiller Replacement Standardization

Program Type (PEO, Enterprise, of Operational): Other Contracting - Air Force Category Management of Cat #4, Facilities and Construction

Authority:

6.302-1 – 10 USC 3204(a)(1), Only One Responsible Source - No Other Supplies or Services Will Satisfy Agency Requirements

Estimated Contract Cost (including options): \$ 437.5M

J&A Type: ☒ Class ☐ Individual

!! Provide estimated cost of all contracts.

COORDINATION (DAFFARS 5306.304(a))

Date 02 Feb 2024	Requiring Activity Clifford M. Theony, Col AFCEC/CO / 850-283-6102	Signature THEONY.CLIFFORD.M.1 030901459	Digitally signed by THEONY.CLIFFORD.M.1030901459 Date: 2024.02.08 16:30:28 -06'00'
Date 04 Mar 2024	Contracting Officer Tommy G. Leos AFICC 772 ESS/PKD / 850-283-6287	Signature LEOS.TOMMY.G.JR.1119 319176	Digitally signed by LEOS.TOMMY.G.JR.1119319176 Date: 2024.03.04 14:52:38 -06'00'
Date 04 Mar 2024	Local Legal Reviewer Karen S. White AFIMSC/JAQ / 850-283-6575	Signature WHITE.KAREN.SUE.1157 422550	Digitally signed by WHITE.KAREN.SUE.1157422550 Date: 2024.03.04 14:58:07 -06'00'
Date 29 Mar 2024	Competition Advocate Michael M. Talbott AFICC/KP / 937-516-2163	Signature TALBOTT.MICHAEL.M.1 258453596	Digitally signed by TALBOTT.MICHAEL.M.1258453596 Date: 2024.03.29 07:55:26 -04'00'
Date 08 Jul 2024	Chief of the Contracting Office (COCO) William F. Barrett, Lt Col AFICC 772 ESS/CC / DSN 969-8776	Signature BARRETT.WILLIAM.FRA NCIS.1277540181	Digitally signed by BARRETT.WILLIAM.FRANCIS.1277540181 Date: 2024.07.08 14:55:05 -05'00'
Date 08 Jul 2024	Senior Contracting Official (SCO) Anthony W. Everidge, SES AFICC/CA / DSN 787-1900	Signature EVERIDGE.ANTHONY.W. 1230167814	Digitally signed by EVERIDGE.ANTHONY.W.1230167814 Date: 2024.07.08 16:31:53 -04'00'

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Justification and Approval (J&A) for Other Than Full and Open Competition

Date	Director AFMC Jeffrey P. Domm, SES AFCEC/CA / DSN 969-8363	Signature DOMM.JEFFREY.P.1146 536189	Digitally signed by DOMM.JEFFREY.P.1146536189 Date: 2024.07.12 10:53:41 -05'00'
Date 23 Jul 2024	Deputy Assistant Secretary (Contracting) Alice W. Trevino, Maj Gen SAF/AQC / 571-256-2397	Signature TREVINO.ALICE.WARD.1 082776300	Digitally signed by TREVINO.ALICE.WARD.1082776300 Date: 2024.07.23 17:58:00 -04'00'

APPROVAL ([DAFFARS 5306.304\(a\)](#))

Date 10SEP2024	Assistant Secretary of the Air Force, Acquisition, Technology, and Logistics Hon Andrew P. Hunter	Signature 
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I. Agency and Contracting Activity.

Department of the Air Force, Air Force Installation Contracting Center, 772 Enterprise Sourcing Squadron (PKD), 139 Barnes Dr., Tyndall AFB, FL 32403

II. Nature and/or description of the action being approved.

The Department of the Air Force (DAF) has a requirement to limit the Heating, Ventilation, & Air Conditioning (HVAC) chiller brand manufacturers on its installations (i.e., brand name) to reduce overall total cost of ownership (TCO). Based on existing HVAC chiller inventory data (i.e., previous investments) and ongoing Air Force Civil Engineer Center (AFCEC) standardization policy compliance, each DAF installation has an approved HVAC chiller manufacturer brand identified. That selected HVAC chiller brand can be procured competitively among authorized distributors that support the installations. The preponderance of these local distributors are small businesses. This Brand Name Class J&A will support multiple DAF installation-level HVAC chiller contracts throughout CONUS and OCONUS locations.

III. Description of supplies/services required to meet agency needs.

This Brand Name Class J&A covers all DAF HVAC chiller purchases from 1 Oct 2024 through 30 Sep 2029. HVAC chiller purchases during the next 5 years are expected to number 1,654 units (i.e., the number of chillers that are least 17 years old by 2029). The value of the HVAC chiller demand is estimated to be \$437.5M based on the replacement value of the units in the government's inventory tracking system, BUILDER. HVAC chillers are mission essential equipment required for regular operations at every DAF installation. They keep critical equipment such as computer servers, aircraft simulators, and facilities with no natural method of ventilation within required temperature and humidity parameters. The strategy of using selected large manufacturers at an installation will not impact the commercial HVAC market. The U.S. HVAC systems market size was estimated at \$30.41 billion in 2023 and is projected to grow at a compound annual growth rate (CAGR) of 7.4% from 2024 to 2030. At approximately \$88M annually, the DAF expenditures for new HVAC systems is 0.29% of the annual U.S. commercial HVAC sales market. By contrast, since standardization, 92% of chiller purchases have been with small businesses, a 19% improvement from 73% under the prior approach.

The purchase of HVAC chillers is a commercial item procurement, and in terms of Intellectual Property (IP), the Government only orders technical data needed for operation and maintenance of the chillers that is customarily provided to the public.

IV. Demonstration that the contractor's unique qualifications or the nature of the acquisition requires use of the authority cited above.

In accordance with FAR 6.302-1(a)(2), the HVAC chillers are available from only one or a limited number of responsible sources and no other type of supplies will satisfy agency requirements. Specifically, as described in FAR 6.302-1(b)(4), the DAF has determined in accordance with its HVAC chiller standardization policy that only the makes of HVAC chiller identified will satisfy the agency's needs for additional units or replacement items for the installations referenced therein. In 2018, the DAF conducted a robust market analysis via a Category Intelligence Report (CIR) which identified relevant acquisition and non-acquisition solutions to optimize HVAC chiller management. Leveraging proven industry best practices, the 2018 CIR determined that standardization to an individual HVAC brand at each installation would reduce the DAF HVAC TCO by reducing training inefficiencies, increasing the responsiveness and effectiveness of in-house maintainers, and lead to a more cost-effective delivery of mission capability. The original Brand Name Class J&A was approved on 07 Aug 2019.

The following findings were identified in the CIR and the 2019 J&A:

1. Practice of selecting chillers solely based on the initial purchases price led to 142 different chiller manufacturers products in the DAF inventory.
2. There are over 3,000 organic DAF HVAC maintainers; 65% of necessary training is unique to each manufacturer's system.
3. Lack of standardization led to inefficient, ineffective, and significant sustainment costs, including training,

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maintenance, and inventory management, which are compounded over the life cycle.

4. Advanced training cannot be delivered virtually.
5. TCO model formula recommended as the basis to determine the total cost of the chiller systems throughout their life cycles but noted the enterprise lacks tools to properly complete a TCO model. Estimated \$50M reduction in training costs for every 5-year training cycle based on cost avoidance.

Throughout this first 5-year period, there were several factors continuing to affect the ability to collect data. The tools to track cost data require significant modifications to align with the TCO components. As an example, BUILDER and NexGen IT data tracks HVAC work, but not the actual cost of individual chillers. Similarly, the systems cannot track costs where chillers are a component of a contract with a larger scope such as a whole building renovation. Data collection was also further hindered during COVID, from 2020 to 2021, during which in-person, advanced training was not available. Also complicating the ability to track savings data is the long life of HVAC Chillers, 17 years on average, as well as the replacement of existing chillers through attrition rather than an established time frame since specific funding is not provided for HVAC chiller replacement. As a result, a complete data analysis will not be seen until 10 to 15 years from the start of the standardization initiative.

AFCEC has made several changes to improve its data gathering capabilities:

1. In January 2024, AFCEC/CO created a Category Management Working Group to help solve the continuity problem by establishing a network within the directorate, building a team to identify solutions to the data collection/analysis challenges, and creating business rules to provide direction to the installations on what data to track and how.
2. Developing a CE DASH tool to process standardization approvals, by automating the request/approval process, storing installation approvals for the installation, and allow for further development for increase future functionality. This tool is expected to be live in October 2024.
3. In February 2024, AFCEC updated NexGen to allow the maintainers to annotate if they worked on a standardization system. This will allow the DAF to track the work performed on a standardized system and will allow the use of data associated with the work order such as labor hours/costs, material costs, service contract costs, and the type of work order (Preventative, Corrective, Emergency).

Despite funding, data system, and other limitations, the DAF has also made progress toward standardization over the past 5 years. Other benefits including cybersecurity, resiliency, and Small Business support have been realized. The total number of manufacturers in the DAF decreased from 142 in 2018 to 127 as of 2024, a reduction of 15 manufacturers. Based on AFCECs continued evaluation of the strategy, this Brand Name Class J&A will expand on the 2019 J&A strategy by allowing a limited number, rather than a single HVAC manufacturer, for each installation. A breakdown of the 87 installations/geographic areas with approved standard chiller manufacturers is provided below:

1. Twelve bases are at least 80% standardized, with the approved standard chillers making up 80% of the base inventory. There are another six installations within 5% of the threshold, however DAF-wide standardization will take 10 to 15 more years given the average 17 year life of the HVAC systems.
2. Eighty-seven installations standardized their HVAC chiller brands since the 2019 J&A resulting in eleven distinct HVAC chiller manufacturers being selected, with two manufacturers representing the standard for 86% of the installations: Trane (57.5%), Carrier (28.7%), Daikin (3.4%), York (2%), LG (1.2%), Mitsubishi (1.2%), Dunham-Bush (1.2%), Exhausto (1.2%), Blue Box (1.2%), Uniflair (1.2%), and United Technology (1.2%).
3. In CONUS there are four Standard Manufacturers, representing 77% of total DAF HVAC chiller inventory: Trane (70%), Carrier (22%), Daikin (5%), and York (3%).
4. In OCONUS there are nine Standard Manufacturers, representing 23% of total DAF HVAC chiller inventory: Carrier (50%), Trane (15%), LG (5%), Mitsubishi (5%), Dunham-Bush (5%), Exhausto (5%), Blue Box (5%), Uniflair (5%), and United Technology (5%).

Benefits achieved thus far from standardization include:

1. Avoided \$50M in Training Costs - In 2021, a new vendor training program was started which included centralized funding for HVAC training for the 2 primary manufacturers. Thus far, 380 maintainer seats have been filled since AFCEC's centrally funded training program began in FY21. If the DAF were to provide this level of support for every

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non-standardized HVAC chiller manufacturer, the advanced training costs would double for each additional manufacturer. There are an average of seven chiller manufacturers on each installation (87). One set of advanced training courses costs approximately \$8k per maintainer. If the DAF funded the same training for the seven manufacturers, it would cost approximately \$56k per maintainer. By standardizing there is an avoidance of \$48k per maintainer, which equates to an estimated DAF-wide cost avoidance of \$50M assuming an average 12-person shop for the 87 installations with approved standardized manufacturers.

2. **Increase in Cybersecurity** - DAF installations are also becoming more cyber secure as they standardize their HVAC chiller inventory. The more contract support is required the more the systems are vulnerable to malicious software being introduced into the system. By standardizing the HVAC chiller brands on the installation, DAF technicians can become experts on the selected equipment, decreasing the need for outsourcing repair services. Additionally, standardization allows the installation to ensure the most cybersecure hardware and software packages are on all the equipment and creates the opportunity to work with the manufacturers on securing vulnerabilities.
3. **Enhanced Resilience with Standardization** - Bases report that by reducing the numbers of systems on the installation, in-house maintainers become experts on those systems improving the diagnostic/troubleshooting, resulting in reduced maintenance, service times and system failures. Additionally, parts from older systems can be utilized to keep current systems running.
4. **Improved Mission Effectiveness and Reliability** - HVAC chiller training is more defined and DAF technicians have become more effective through equipment familiarity. Recent installation feedback indicates HVAC chiller diagnostics and troubleshooting are much easier for technicians, resulting in reduced maintenance and service time with fewer system failures.
5. **Reduced Equipment and Parts Inventories** - DAF installations have reported reductions in spare part inventory demand due to lower HVAC chiller inventory variation (i.e., fewer brands per installation versus up to 17 different brands prior to standardization). Standardization allows DAF to interchange many parts, increasing resiliency. When supply chain issues intensified during COVID, decommissioned chillers were available after facility demolition and the work center was able to salvage parts from these units to repair similar HVAC chillers still in use.
6. **Benefits To Small Business and Competition** - The distributors that supply/install the equipment and parts to the installations are typically small businesses. Since standardization, 92% of chiller purchases have been with small businesses. This is a 19% improvement (73% to 92%) since the 2018 CIR. Additionally, there is robust competition amongst the distributors. Before the 2019 class brand name J&A, the DAF averaged 2.875 proposals per requirement based on the PSC/NAICS for HVAC systems. In FY23, this increased to 4.86 proposals and exceeded other Federal organizations, which averaged 3.5 proposals per requirement in FY23.

As outlined above, award to any other sources than those brand names specified for each DAF installation would result in substantial duplication of cost to the Government that is not expected to be recovered through competition.

V. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by subpart 5.2 and, if not, which exception under FAR 5.202 applies.

IAW FAR 5.201 a Request for Information/Sources Sought, 238220, HVAC Chillers, was posted on SAM.Gov on 12 December 2023 and remained open for 15 days.

A brief description of the requirement was included in the notice. The notice explained that this effort was a continuation of the previous efforts, that began in 2019, to standardize HVAC procurement, to include repair and maintenance, across the DAF and allow each DAF installation within and outside the contiguous United States to select, based on its existing inventory, a standard brand of HVAC chiller when new or replacement HVAC chillers are required.

The notice advised the DAF intends to renew the J&A for another five years (FY25 through FY29) to continue standardization of its HVAC chiller inventory through natural attrition and will support various installation-level contracts for new or replacement chillers. The notice asked industry to respond so the DAF could better understand how the planned J&A may

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impact industry.

Two responses from industry were received stating that chiller standardization is an industry best practice that directly benefits the customer by reducing training costs for technicians, enabling purchases of consumables at scale and streamlining maintenance requirements. There were no industry comments received with concerns or negative feedback on the AFCEC acquisition strategy to select a single manufacturer for each installation.

VI. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

Contracting Activities will be required to make individual determinations of price fair and reasonableness IAW FAR 13.106-3 (a) (simplified acquisitions) or FAR 15.403-3(c)(1) (commercial acquisitions) prior to award. AFCEC and local installations have a large database of historical prices based on competitive acquisitions, which will be used for comparisons and for the Government's estimate. While there would not be competition among manufacturers for installation-level purchases of the assigned brand name HVAC chillers under this J&A, costs will still be competed among the local authorized distributors. As previously mentioned, the preponderance of these distributors are small businesses (92%).

VII. Description of the market research conducted and the results, or a statement of the reasons market research was not conducted.

As referenced in section IV, a CIR was executed in 2018. As part of the standard CIR process, the DAF HVAC CIR conducted comprehensive requirements, market, and gap analyses to develop targeted recommendations for improving DAF HVAC chiller management. The CIR's detailed requirements analysis outlined historical DAF contract spend and inventory data as well as internal DAF maintenance, sustainment, and management processes to establish a clear baseline for strategy comparisons. The CIR's thorough market analysis identified external HVAC chiller benchmarks, best practices, and trends from relevant analogous organizations such as universities and airports. Supporting facts in this Brand Name Class J&A were drawn from the aforementioned CIR including the anticipated total cost of ownership reductions associated with HVAC chiller standardization. Furthermore, an RFI/Sources Sought was posted 12 Dec 2023 for 15 days to gather feedback from industry with 2 responses from industry. Those are detailed above in Section V.

VIII. Any other facts supporting the use of Other Than Full and Open Competition.

Market analysis indicates there have not been any significant changes in the HVAC market since the last J&A which was confirmed in the response received to the most recent RFI. In order to increase mission performance and cost savings, reduce cyber threats and innovation in business methods, HVAC chiller standardization supports the National Defense Strategy objective to innovate business decisions for saving taxpayer dollars, which has already yielded an estimated \$50M in cost avoidance for training. In order to implement this management strategy, the DAF must continue to pre-plan the brand name HVAC chillers at each installation.

Since execution of the 2019 J&A, AFCEC has continued to research the HVAC chiller market and work with DAF-wide installations to standardize manufacturers. Installations research the various HVAC manufacturers at their location and provide AFCEC with information to support the desired standardized manufacturer. In many cases, the selection is based on the predominant HVAC chiller at that location, however in some instances other criteria is used to select a standardized manufacturer. As an example, one installation selected their standardized manufacturer based on the quantity and close proximity of the manufacturer distributors to the installation. Benefits from this approach include robust competition for HVAC chiller related needs, use of local, often small businesses, as well as increased availability and speed to obtain necessary parts and support.

IX. List of any sources that expressed, in writing, an interest in the acquisition.

This Brand Name Class J&A does not have a specific acquisition attached to this request. Sources that expressed interest in this effort initially, in 2019, include: Carrier, The York Brand of Johnson Controls, McQuay Chillers of Daikin Applied, Matthis Construction Company, and US Comfort Building Services Inc. Two companies responded to the recent Sources Sought notice as indicated in Section V above. The underlying analysis of the need for standardization accounted for as many different manufacturer's brands as would be practical to achieve the maximum amount of savings and efficiencies. An "or

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equal" manufacturer does not achieve standardization. The efficiencies of maintaining a limited number of HVAC chiller brands over time could not be achieved through an "or equal" approach.

X. A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisitions for the supplies or services required.

The only barriers to competition under this Brand Name Class J&A have been generated by historical purchases. This Brand Name Class J&A action only recognizes that past competitive awards established the number and mix of HVAC chillers on each installation. The Category Management CIR analysis recognized these facts and established a strategy to maximize efficiencies based on what the DAF had in its current inventory across the enterprise and at local levels. The DAF will continue to perform data driven analysis over time to validate the HVAC standardization strategy continues to achieve intended mission outcomes and cost improvements. Additionally, as standard HVAC industry practice involves manufacturers selling through authorized dealers and distributors, competition will be maintained at the dealer/distributor level.

XI. Certification by the Contracting Officer.

As evidenced by my signature above, I certify that this justification is accurate and complete to the best of my knowledge and belief.

XII. Certification by the technical/requirements personnel.

As evidenced by my (our) signature(s) above, I (we) certify that any supporting data contained herein, which is my (our) responsibility, is both accurate and complete.